
COP4600:**Operating Systems**

CRN 15526/15527, Section #001/#002, 3 Credit Hours

COURSE SYLLABUS

Semester: Spring 2025

Class Meeting Days: T, R

Class Meeting Time: 06:30pm-07:45pm

Class Meeting Location: CHE 103 (#001) / CPR 115 (#002)

Instructor: Mauricio Pamplona Segundo

Office Location: ENB 343H

Office Hours: Check Canvas

Email: mauriciop@usf.edu

I. University Course Description

Introduction to systems programming. Design of operating systems. Concurrent processing, synchronization, and storage management policies.

II. Course Prerequisites

Data Structures (COP 4530)

III. Student Learning Outcomes

Upon successful completion of this course, you should be able to:

- Explain the objectives and functions of modern operating systems.
- Describe the basic principles in the design of contemporary operating systems.
- Describe how computing resources (such as CPU and memory) are managed by the operating system.
- Compare and contrast the common algorithms used for both pre-emptive and non-pre-emptive scheduling of tasks in operating systems, such as priority, performance comparison, and fair-share schemes.
- Summarize techniques for achieving synchronization in an operating system.
- Explain memory hierarchy and cost-performance trade-offs.
- Evaluate and report appropriate design choices when solving real-world problems.
- Analyze the key trade-offs between multiple approaches to operating system design.

IV. Course Objectives

- Understand the principles of modern operating system design.
- Recognize and appreciate the differences among the most common operating systems.
- Understand concurrency.

- Learn scheduling techniques.
- Understand main memory management.
- Understand virtual memory management. Be able to evaluate cost and benefits.
- Learn the principles of file management and disk management.

V. Required Texts and/or Readings and Course Materials

- Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Operating Systems: Three Easy Pieces, Arpaci-Dusseau Books, version 1.10, 2023, ISBN-13: 978-1985086593. Available at <https://pages.cs.wisc.edu/~remzi/OSTEP/>

VI. Grading Scale

Grading Scale (%)	
90-100	A
80 - 89	B
70 - 79	C
60 - 69	D
0 - 59	F

VII. Grade Categories and Weights

Graded Items	Percent of Final Grade
Attendance quizzes	5%
Projects	25%
Midterms	40%
Final Exam	30%

* Extra Credit (0.5 pt): Students can earn up to 0.5 extra credit. Please refer to the extra credit policy.

VIII. Course Schedule.

Week 1 (1/14, 1/16): Introduction to operating systems concepts and hardware virtualization
 Week 2 (1/21, 1/23): CPU virtualization and multi-process programming
 Week 3 (1/28, 1/30): Direct execution and CPU scheduling
 Week 4 (2/4, 2/6): Memory virtualization and paging
 Week 5 (2/11, 2/13): **Virtualization review, Midterm #1 (Saturday 2/15), Project #1 due**
 Week 6 (2/18, 2/20): Introduction to concurrency and threads
 Week 7 (2/25, 2/27): Locks, conditions, and the producer-consumer problem
 Week 8 (3/4, 3/6): Semaphores and the readers-writers problem
 Week 9 (3/11, 3/13): Deadlocks and other concurrency bugs
 Week 10 (3/25, 3/27): **Concurrency review, Midterm #2 (Saturday 3/29), Project #2 due**
 Week 11 (4/1, 4/3): Introduction to persistence and IO devices
 Week 12 (4/8, 4/10): Hard disk drives and RAID
 Week 13 (4/15, 4/17): File system implementation
 Week 14 (4/22, 4/24): FSCK and Flash-based SSDs
 Week 15 (4/28, 5/1): **Course wrap-up and review, Project #3 due**
 Week 16 (check final exam matrix): **Cumulative final exam**

* Note: The schedule is subject to revision.

IX. USF Core Syllabus Policies

USF has a set of central policies related to student recording class sessions, academic integrity and grievances, student accessibility services, academic disruption, religious observances, academic continuity, food insecurity, pregnancy and related conditions, and sexual harassment that **apply to all courses at USF**. Be sure to review these online: usf.edu/provost/faculty-success/resources-policies-forms/core-syllabus-policy-statements.aspx

X. Course Policies: Grades

Late Work Policy: Projects turned in late will be assessed a penalty: 20% deduction per day. If you are five days late or more, there is no point in submitting your work.

Medical Excuses: Project descriptions will be available for at least two weeks before the due date, and submissions can be made at any time with real-time feedback through GradeScope. Medical excuses will only be accepted if they cover at least 75% of the period a project is available. If you get sick a day before a deadline and request an extension, this will not be considered a valid request. You must submit verifiable documentation from a medical institution declaring that you were unable to work on a project for at least 75% of the period that it was available to receive an extension.

Extra Credit Policy: Exams may contain challenging questions that will be worth extra credit. There is 0.5 extra credit given for participating in the course evaluation before the course ends. The feedback and evaluation are anonymous, so you will need to upload a screenshot as proof that you have completed it. Extra credits will be added after all grades are entered on Canvas.

Grades of "Incomplete": The current university policy concerning incomplete grades will be followed in this course.

Attendance Policy: Attendance quizzes will be given at random lectures. 25% of attendance quiz grades will be dropped. There is no makeup for attendance quizzes.

Make-up Exams Policy: If a student cannot be present for an examination for a valid reason (validity to be determined by the instructor), a make-up exam will be given only if the student has notified the instructor in advance that s/he cannot be present for the exam. Make-up exams are given at the convenience of the instructor.

Group Work Policy: At least one project will be a group project. Groups will be created randomly, and the grades of individuals within the same group may differ.

Final Examinations Policy: All final exams are to be scheduled in accordance with the University's final examination policy.

XI. Course Policies: Technology and Media

GradeScope: This course uses GradeScope for grading projects automatically: <https://www.gradescope.com/>

XII. Course Policies: Student Expectations

Health and Wellness: Your health is a priority at the University of South Florida. We encourage members of our community to look out for each other and to reach out for help if someone is in need. If you or someone you know is in distress, please make a referral at www.usf.edu/sos so that the Student Outreach & Support can contact and provide helpful resources to the student in distress. A 24-hour licensed mental healthcare professional, offered through the counseling center, is available by phone at 813-974-2831, option 3. Please remember that asking for help is a sign of strength. In case of emergency, please dial 9-1-1.

Title IX Policy: Title IX provides federal protections for discrimination based on sex, which includes discrimination based on pregnancy, sexual harassment, and interpersonal violence. In an effort to provide support and equal access, **USF has designated all faculty (TA, Adjunct, etc.) as Responsible Employees, who are required to report any disclosures of sexual harassment, sexual violence, relationship violence or stalking.** The Title IX Office makes every effort, when safe to do so, to reach out and provide resources and accommodations, and to discuss possible options for resolution. Anyone wishing to make a Title IX report or seeking accommodations may do so online, in person, via phone, or email to the Title IX Office. For information about Title IX or for a full list of resources please visit: <https://www.usf.edu/title-ix/gethelp/resources.aspx>. *If you are unsure what to do, please contact Victim Advocacy – a confidential resource that can review all your options – at 813-974-5756 [or va@admin.usf.edu](mailto:va@admin.usf.edu).*

Generative AI, Plagiarism, and Cheating: The use of Generative AI for coding or coding assistance is not allowed in this course and is considered cheating. Other uses, such as text and illustration generation for presentations, are allowed. Any form of cheating will be penalized with a failing grade in the course.

XIII. Learning Support and Campus Offices

Academic Accommodations

Students with disabilities are responsible for registering with Student Accessibility Services (SAS) in order to receive academic accommodations. For additional information about academic accommodations and resources, you can visit the SAS website.

[SAS website for the Tampa and Sarasota-Manatee campuses.](#)

Academic Support Services

The USF Office of Student Success coordinates and promotes university-wide efforts to enhance undergraduate and graduate student success. For a comprehensive list of academic support services available to all USF students, please visit the [Office of Student Success website](#).

Canvas Technical Support

Include information where students can find technical support.

Example: If you have technical difficulties in Canvas, you can find access to the Canvas guides and video resources in the “Canvas Help” page on the homepage of your Canvas course. You can also contact the help desk by calling 813-974-1222 in Tampa or emailing help@usf.edu.

[IT website for the Tampa campus.](#)

Center for Victim Advocacy

Example: The [Center for Victim Advocacy](#) empowers survivors of crime, violence, or abuse by promoting the restoration of decision making, by advocating for their rights, and by

offering support and resources. Contact information is available online.

Counseling Center

Example: The Counseling Center promotes the wellbeing of the campus community by providing culturally sensitive counseling, consultation, prevention, and training that enhances student academic and personal success. Contact information is available online.

[Counseling Center website for the Tampa campus.](#)

Tutoring

Example: The Tutoring Hub offers free tutoring in several subjects to USF undergraduates. Appointments are recommended, but not required. For more information, email

[asctampa@usf.edu.](mailto:asctampa@usf.edu)

[Tutoring website for the Tampa campus.](#)

XIV. Important Dates to Remember

<i>Drop/Add Deadline:</i>	<i>Fri, 01/17/25</i>
<i>Dr. Martin Luther King, Jr. Holiday:</i>	<i>Mon, 01/20/25</i>
<i>Mid-term Grading Opens:</i>	<i>Mon, 02/17/25</i>
<i>Mid-term Grading Closes:</i>	<i>Tues, 03/04/25</i>
<i>Spring Break</i>	<i>03/17/25 – 03/23/25</i>
<i>Withdrawal Deadline:</i>	<i>Sat, 03/29/25</i>
<i>Final Examination Week:</i>	<i>05/03/25 – 05/08/25</i>